ATTACHING A SINGLE NONDEGENERATE SIMPLEX GIVES A PUSHOUT DIAGRAM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the induction on degeneracies (source lines 3330-3336) is the engine of surjectivity; the classification of nondegenerate simplices of the pushout (source lines 3338-3341) is the point of injectivity; no verification is omitted.

Editorial note. The label lemma-glue-simplex names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Simplicial subsets and the skeleton attachment. Statement lines 3300-3318. U subset V simplicial sets with V_i = U_i for i < n, V_n = U_n + {x}, and all simplices outside U in degrees > n degenerate; the boundary i_{(n-1)!} sk_{n-1} Delta[n] = partial Delta[n].
- Injectivity of maps of simplicial sets via nondegenerate simplices. Same chapter, \label{lemma-injective-map-simplicial-sets}. Proof lines 3440-3444 region of the cited lemma: a map of simplicial sets injective on nondegenerate simplices and mapping nondegenerate to nondegenerate is injective; used to conclude.

Source reference. Original reference 017S: sources/stacks-simplicial/pinned-original-source.tex, source lines 2041–2041. Prerequisite pointer invoked at proof line(s) 3342; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $U \subset V$ be simplicial sets. Suppose $n \geq 0$ and $x \in V_n$, $x \notin U_n$ are such that*

(1) *$V_i = U_i$ for $i < n$,*
(2) *$V_n = U_n \cup \{x\}$,*
(3) *any $z \in V_j$, $z \notin U_j$ for $j > n$ is degenerate.*

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Let $\Delta[n] \to V$ be the unique morphism mapping the nondegenerate n-simplex of $\Delta[n]$ to x. In this case the diagram

$$\begin{array}{ccc} \Delta[n] & \longrightarrow & V \\ \uparrow & & \uparrow \\ i_{(n-1)!}sk_{n-1}\Delta[n] & \longrightarrow & U \end{array}$$

is a pushout diagram.

Proof. Let us denote $\partial\Delta[n] = i_{(n-1)!}\mathrm{sk}_{n-1}\Delta[n]$ for convenience. There is a natural map $U \amalg_{\partial\Delta[n]} \Delta[n] \to V$. We have to show that it is bijective in degree j for all j. This is clear for $j \leq n$. Let $j > n$. The third condition means that any $z \in V_j$, $z \notin U_j$ is a degenerate simplex, say $z = s_i^{j-1}(z')$. Of course $z' \notin U_{j-1}$. By induction it follows that z' is a degeneracy of x. Thus we conclude that all j-simplices of V are either in U or degeneracies of x. This implies that the map $U\amalg_{\partial\Delta[n]}\Delta[n] \to V$ is surjective. Note that a nondegenerate simplex of $U\amalg_{\partial\Delta[n]}\Delta[n]$ is either the image of a nondegenerate simplex of U, or the image of the (unique) nondegenerate n-simplex of $\Delta[n]$. Since clearly x is nondegenerate we deduce that $U \amalg_{\partial\Delta[n]} \Delta[n] \to V$ maps nondegenerate simplices to nondegenerate simplices and is injective on nondegenerate simplices. Hence it is injective, by Lemma 017S. $\square$